

REACHING THE NOVICE OR NUDGING THE EXPERT? NETWORKS, INFORMATION, AND EXPERIMENTAL RETURNS TO MIGRATION

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AddRan Research Spotlight

THANKS AND DISCLAIMERS



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STARK URBAN-RURAL GAP IN KENYA



Pictures show a rural Kenyan home and Nairobi, the capital city of Kenya. Source: Jocelyn Diaz (left) and Catalin Marin (right).

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- urbanization is limited
 - Nairobi: **9%** of Kenya's population
 - the rest mostly rural and agricultural
 - urbanization is key to development: transition from agriculture to industry/services

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- **prior literature:**
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 - avg: 45%
 - because experimenting is costly, migrant relatives hide income

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- **sample:** large population-representative sample of 5 Kenyan counties

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 - should target the inexperienced individually, help form new **destination** links

EXPERIMENT DESIGN

SAMPLING

- select 5 counties representative of rural Kenya ► *county sampling*
 - sample is mostly typical of rural Kenyan population, but poorer ► *comparison*

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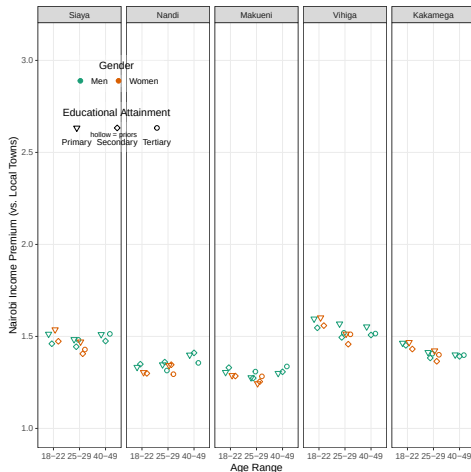
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 - sample 30 households per village for RCT
 - ~30% of avg village size
 - 16,878 households sampled
- *household sampling*

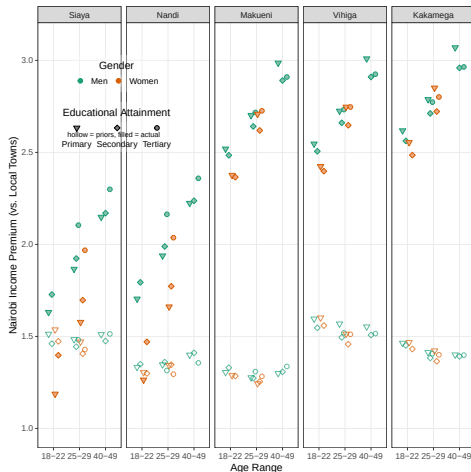
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- rural households underestimate conditional Nairobi income by **45%** on average
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- most HHs have **limited migration networks**:
 - 56% have no member with Nairobi work experience
 - 54% have no one to ask about jobs in Nairobi
 - believe that 16% of the village has a member working in Nairobi (reality: 29%)
 - households with migration experience score better on all these measures
 - *urban networks*

INTERVENTION: VILLAGES RANDOMLY ASSIGNED TO TREATMENTS

- information

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- information
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- control

INFORMATION TREATMENT

- **information** treatment: information brochure + script ► *details*
 - avg income, income distribution in Nairobi by demographic group
 - employment rates
 - rental prices and amenities
 - time for Q&A with staff

INCOME IN NAIROBI

WHO?


Men ages
25-29
with Form 4

HOW MUCH? (KSh per month)

TYPICAL

25,700

LOW

HIGH

13,500 – 31,500

WORKING

91 out of 100

COMPARED TO towns in Kakamega

That's

3

times as much as in
Kakamega towns


Women ages
25-29
with Form 4

TYPICAL

17,800

LOW

HIGH

10,200 – 20,700

WORKING

72 out of 100

That's

2

times as much as in
Kakamega towns

How much is
rent in Nairobi?



ONE ROOM (KSh per month)

TYPICAL

4,000

LOW

HIGH

2,900–5,400

TWO ROOMS (KSh per month)

TYPICAL

10,000

LOW

HIGH

5,700–12,800

A TYPICAL EXPERIENCE

John is **27** years old
and lives in Nairobi. He
graduated from Form 4

▶ He earns
about **25,000**
per month

▶ He lives in a **1-room**
home and pays
4,300 per month

▶ He has a kerosene stove,
water piped to plot, a flush
toilet, and electricity

What
about other
groups?

- Men ages 18–22 with Std 8 typically earn 20,800 per month
- Women ages 18–22 with Std 8 typically earn 12,900 per month
- Men ages 40–59 with Form 4 typically earn 30,700 per month
- Women ages 40–59 with Form 4 typically earn 22,900 per month
- Fewer 18–22 year-olds are working
53 out of 100 for men | 33 out of 100 for women

What do these
numbers mean?



Poorest



Richest

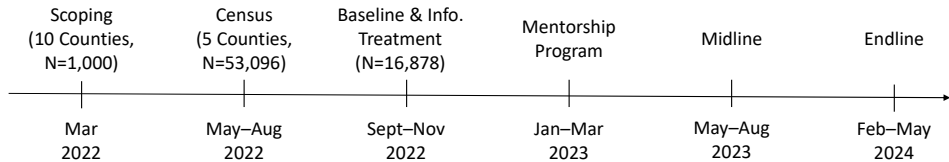
MENTOR TREATMENT

- **mentor** treatment: same info + match to urban mentor ► *details*
 - same brochure + script
 - matched with urban mentor for 1-on-1 support
 - mentors recruited from long-term migrants in Nairobi
 - encouraged to speak over the phone, meet in person ► *about what?*

GROUP TREATMENT

- **group** treatment: same info + group meeting ► *details*
 - village-wide meeting of treated HHs
 - same info brochure + script
 - staff facilitate group discussion, inviting prior migrants to talk, answer questions
 - break into small groups to discuss migration plans

TIMELINE



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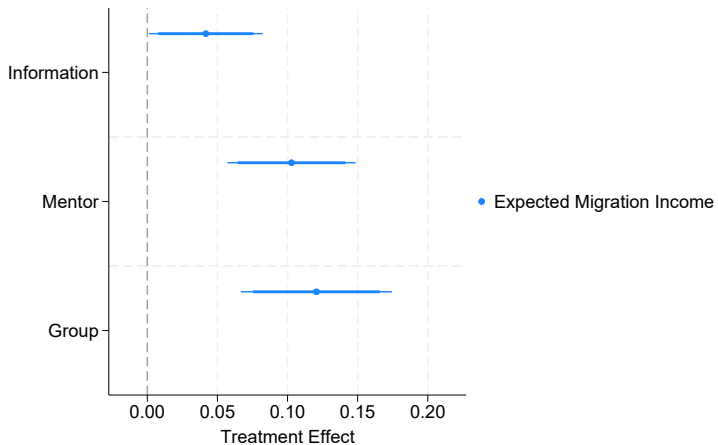
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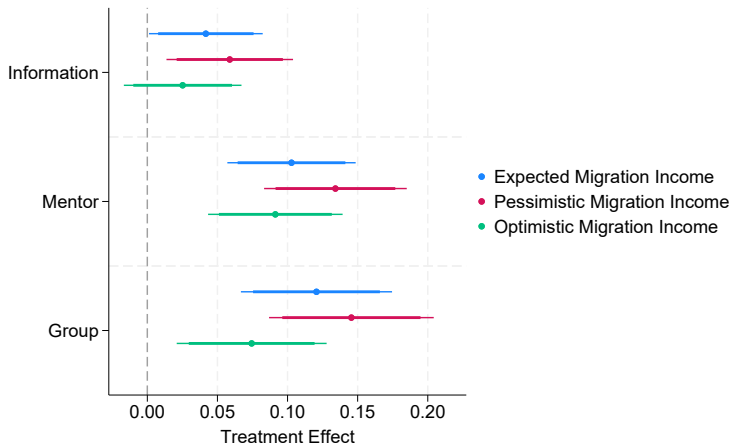
EXPERIMENTAL IMPACTS

BETTER INFO → PERCEIVED URBAN INCOME ↑, MIGRATION PLANS ↑



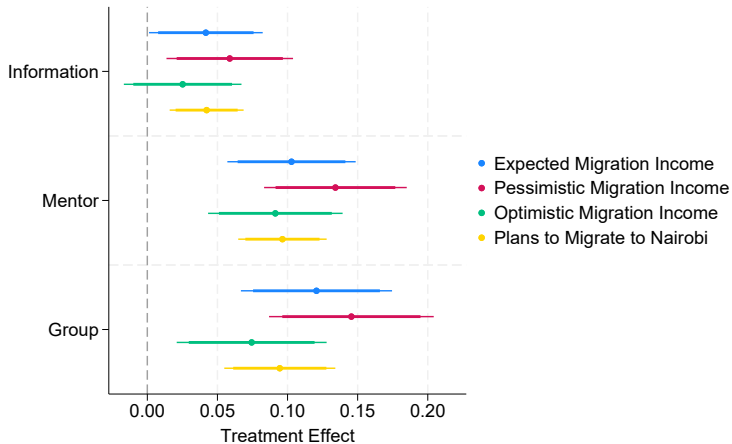
Measured at baseline, after info provision. Units are percentage points (migration plans) or % changes (income beliefs). Bars show 90 and 95% confidence intervals.

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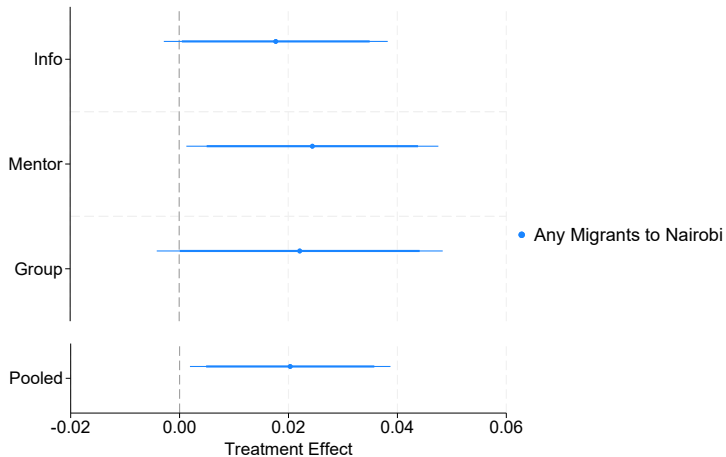
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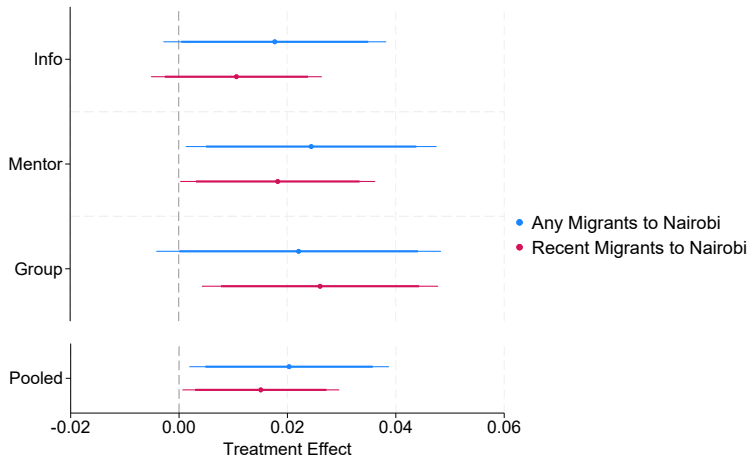
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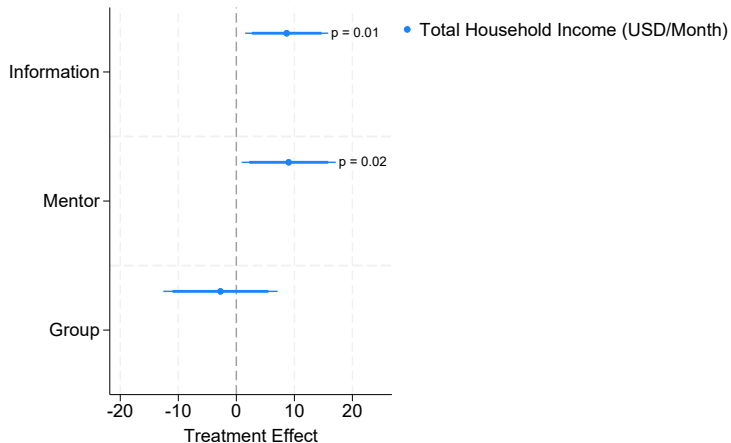
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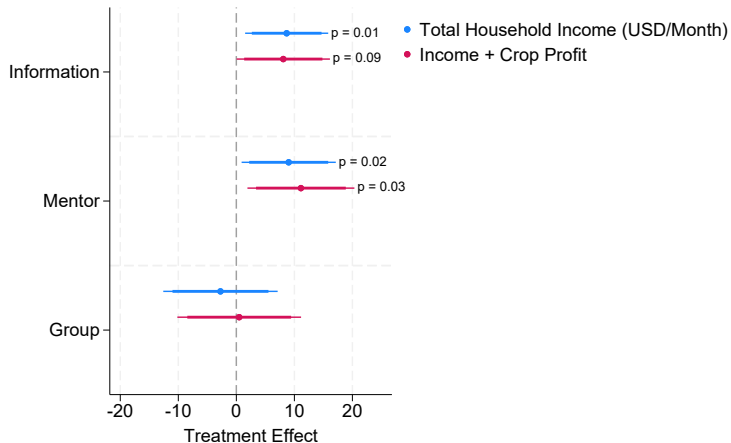
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► *prices* ► *amenities* ► *other outcomes*

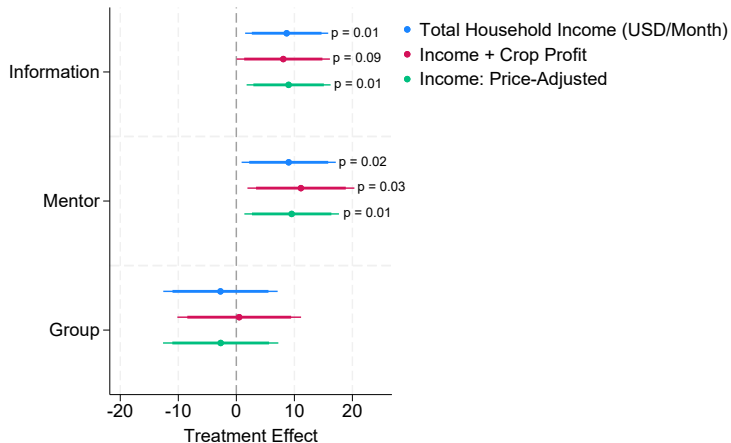
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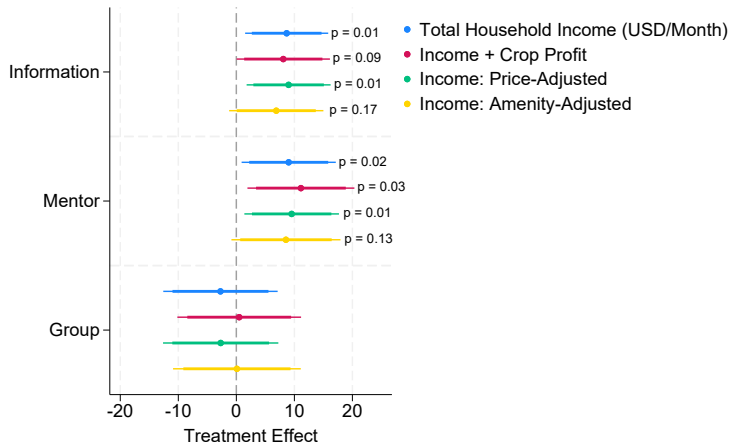
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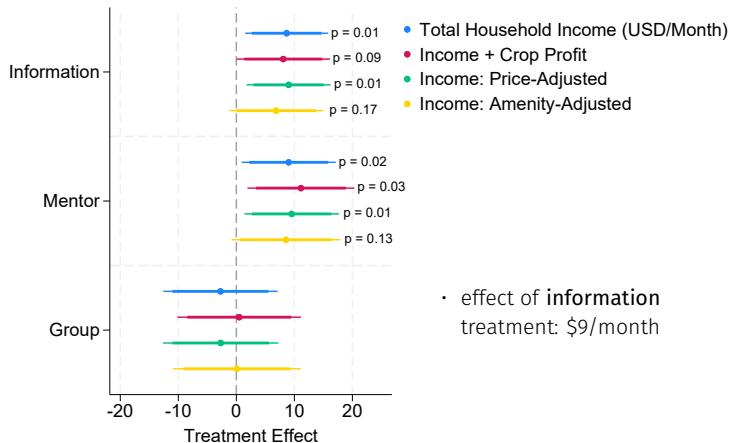
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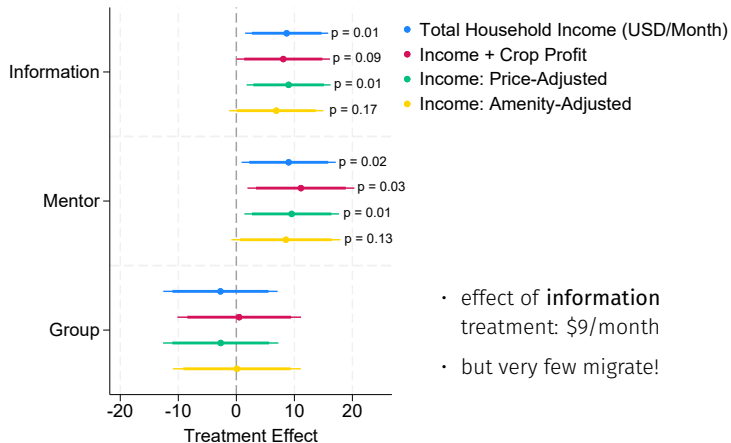


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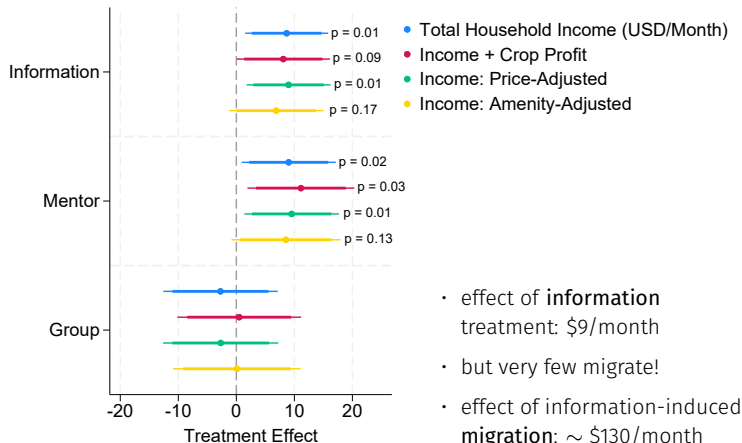


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- → economic returns to **group** treatment are lower

MODEL

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 - **fix $\frac{1}{4}$ of the $5.1\times$ urban-rural income gap**

CONCLUSION

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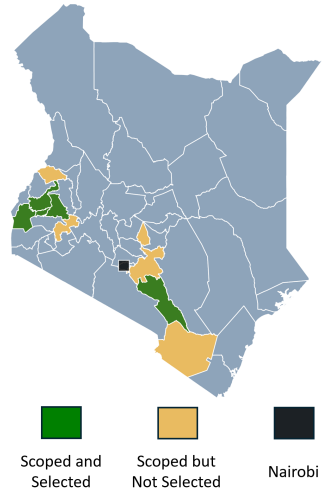
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 - experienced low-return migrants dominate group discussion
- to “reach the novice”,
 - deliver info individually
 - help form network connections “at destination”

STEP 1: COUNTY SELECTION

- Choose 10 counties based on census data, balancing heterogeneity and representativeness
- Short “scoping” surveys in random villages across 10 counties
- Narrow down to five, prioritizing counties with moderate baseline migration



[back](#)

SAMPLE IS TYPICAL OF RURAL KENYAN POPULATION

	Sample Counties	All Counties	Percentile
% aged 18–50	0.36	0.37	0.41
% with primary degree	0.38	0.38	0.55
% with secondary degree	0.07	0.08	0.39
% with post-secondary degree	0.01	0.02	0.39
% Muslim	0.01	0.00	0.73
Per-capita household income (USD/month)	13.57	23.81	0.26
Density (pop. per sq. km.)	393	288	0.67
Distance to Nairobi (km)	393	306	0.73
% of households migrated out of county	0.23	0.20	0.76
Population	3,972,090	26,384,420	

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STEP 2: VILLAGE SELECTION AND HOUSEHOLD CENSUS

- Universe of villages from Kenya National Bureau of Statistics
- Randomly select sub-locations (clusters of 10 villages)
 - Exclude bottom 5% and top 10% of county-level population density
- Randomly select one village per sub-location
 - Reduces inter-village spillover risk
 - Exclude villages with < 50 households
- Census entire village $\implies$ sample of 53,096 households
 - Found 102 households per village (admin data shows 99)
 - Surveyed 90% of all households, sampling weights account for survey probability

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STEP 3: HOUSEHOLD SELECTION AND RCT

- Randomly select 30 households per village $\implies$ RCT sample of 16,878 households
 - Stratify on intended migration, oversampling likely migrants (re-weight in estimation)
- Village-level treatment randomization
 - Stratify by county, share intending to migrate to Nairobi, average income
 - Balanced: 3/31 F -tests sig. at 10%, χ^2 -test with RI (Kerwin, Rostom, Sterck 2024)
- 8-month phone midline: 81% completion, not differential
- 16-month in-person endline: 95% completion, not differential
- Migrant phone surveys: 86% completion, not differential

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INTERVENTION DETAILS: INFORMATION

- Total individual income in Nairobi by age, gender, education
 - Wages + (enterprise profit) / # entrepreneurs + (crop profit + other income) / # adults
 - Three quantiles: p25, p50, p75
 - Median incomes relative to a reference point (towns in the home county)
- Employment rates (working 20+ hours in a typical week)
- Rental prices and typical housing amenities
- Source: Kenya Integrated Household Budget Survey 2015–2016
- Treated households get information brochure + script
 - Built-in time for questions, back-and-forth with staff

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INTERVENTION DETAILS: GROUP

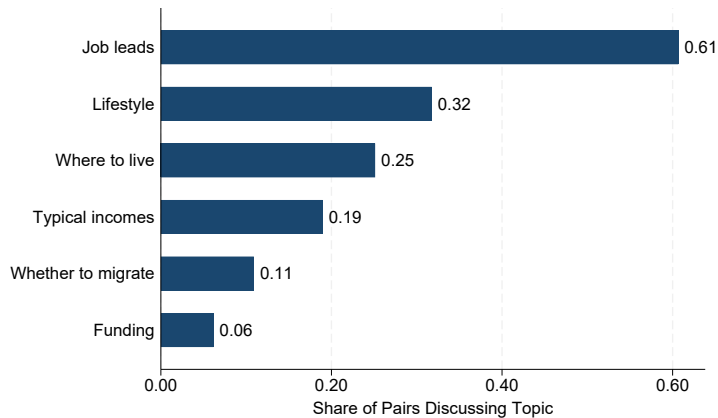
- Each sampled household invited to a group presentation, one or two per village
- Everyone receives the same info sheet, staff reads the same script
- Staff facilitated group discussions about migrating, inviting prior migrants to talk + take questions
- Break into small groups to discuss migrating or coordinating trips
- Attendance was 88% of invited sample
- For those not attending, give individual version during a follow-up visit

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INTERVENTION DETAILS: MENTOR

- Basic info sheet and script + offered a 1-on-1 “mentor”
 - We recruited mentors from popular migrant neighborhoods throughout Nairobi
 - Screened on experience (lived in Nairobi for 5+ years)
 - Mentors agreed to speak with them over the phone and/or meet them in Nairobi
 - Mentors received 500 KSh (\$5) per meeting
- Matching was done live as participants enrolled [Matching Details](#)
 - Collected preferences from villagers and characteristics from mentors
 - “Greedy matching”: we gave people the best available match once they enrolled
- Mentors available starting in January 2023 (two months after baseline), program was open for three months.
- 471 households (13% of the sample) participated (staff verified). Of these, 41 physically met in Nairobi.

WHAT DID MENTORS TALK ABOUT?



Data from midline surveys of household heads.

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ASSESSING BASELINE INFORMATION GAPS

- Elicit income beliefs about Nairobi and reference towns (towns in home county) during census surveys ($N = 53,096$)
- Compare to estimates from representative surveys
- Ask belief about specific age-by-gender cells, randomized across households
- Each household was asked the same question for primary school, secondary school, and college graduation
- Compute “Nairobi premium” for group g as the mean belief about Nairobi over the mean belief about the reference town

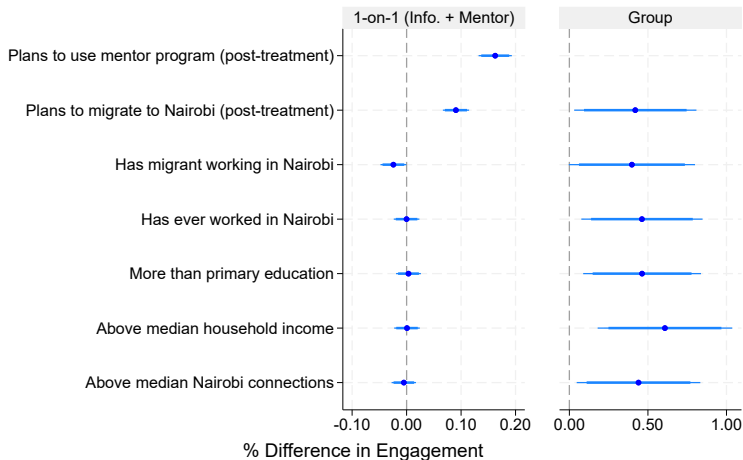
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WHY DIDN'T INFORMATION DIFFUSE?

- Recall that the average household underestimates the share of its village with a migrant by almost half
- Are there strategic incentives to conceal the information given during treatment?
 - Informing others may make them more likely to co-migrate...
 - But it may become harder to hide migration status or income
- We find that households with more potential creditors in the village update less in Spillover (vs. Pure Control) villages [Table](#)
 - No differences for non-financial relationships
 - Helps explain the success of mentors: they are outside rural households' risk-sharing networks

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CHANNEL 1: INEXPERIENCED MIGRANTS ARE LESS ENGAGED IN GROUP TREATMENTS, BUT NOT 1-ON-1 TREATMENTS*



WHEN EXPERIENCED MIGRANTS LEAD GROUP DISCUSSIONS, THEY PRIMARILY INFLUENCE OTHER EXPERIENCED MIGRANTS*

Outcome: Recent Migrants to Nairobi	Measure of Experience, X		
	Has Migrant in Nairobi	Has Worker in Nairobi	Ever Worked in Nairobi
Leader of Type X $\times$ X $\times$ Group	0.13*** (0.03)	0.05* (0.03)	0.06* (0.03)
Leader of Type X $\times$ Group	-0.02 (0.02)	0.01 (0.02)	-0.00 (0.02)
X $\times$ Group	0.07*** (0.01)	0.06*** (0.01)	0.03*** (0.01)
Group	0.00 (0.01)	0.01 (0.01)	0.01 (0.01)
Group Mean (X)	0.36	0.29	0.43
Group Mean (Leader of Type X)	0.71	0.62	0.72
Observations	15,468	15,468	15,468

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CHANNEL 2: INFO. TREATMENT FACILITATES DIS-ASSORTATIVE CO-MIGRATION*

	Co-Migrated to Nairobi		Co-Migrated to Nairobi With More Experienced Migrant	
Info	0.006***	0.007***	0.006***	0.007***
	(0.002)	(0.002)	(0.002)	(0.002)
Group	0.004	0.002	0.002	0.002
	(0.003)	(0.002)	(0.002)	(0.001)
Mentor	0.001	0.003	0.001	0.002
	(0.002)	(0.002)	(0.002)	(0.001)
Info × Mig. In Nairobi		-0.005		-0.004
		(0.005)		(0.004)
Group × Mig. In Nairobi		0.008		0.003
		(0.009)		(0.008)
Mentor × Mig. In Nairobi		-0.006		-0.003
		(0.004)		(0.003)
Control Mean	0.008	0.008	0.006	0.006
Observations	15,468	15,468	15,468	15,468

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Group	0.004 (0.003)	0.002 (0.002)	0.002 (0.002)	0.002 (0.001)
Mentor	0.001 (0.002)	0.003 (0.002)	0.001 (0.002)	0.002 (0.001)
Info × Mig. In Nairobi		-0.005 (0.005)		-0.004 (0.004)
Group × Mig. In Nairobi		0.008 (0.009)		0.003 (0.008)
Mentor × Mig. In Nairobi		-0.006 (0.004)		-0.003 (0.003)
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CHANNEL 2: INFO. TREATMENT FACILITATES DIS-ASSORTATIVE CO-MIGRATION*

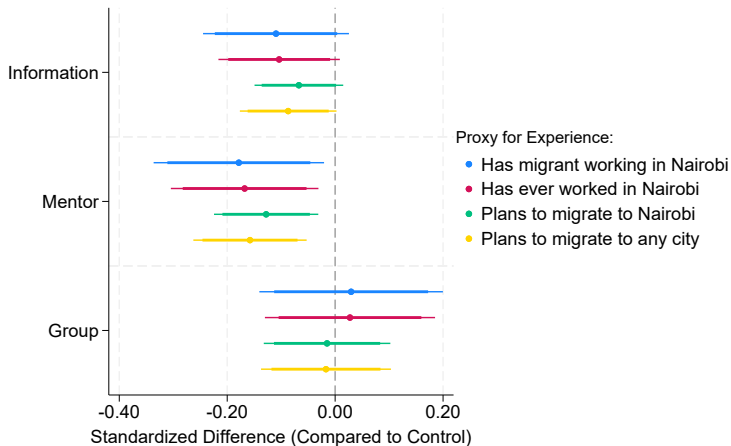
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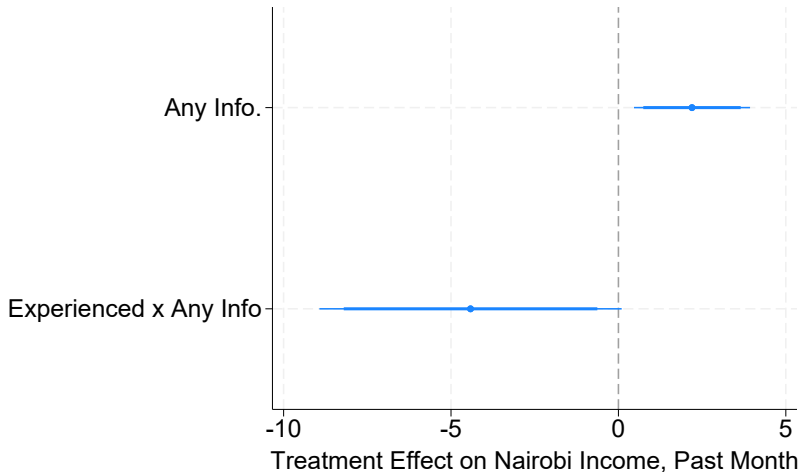
INFO. AND MENTOR INDUCE LOW-EXPERIENCE MIGRATION COMPARED TO GROUP*



Each outcome is a baseline variable. Sample includes households with migrants at endline.

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INEXPERIENCED MIGRANTS HAVE HIGHER RETURNS AT THE MARGIN



Any Info is a pooled treatment variable. *Experienced* = 1 if the hh had a migrant working in Nairobi before treatment.

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CONSUMER PROBLEM

$$\max_{c_i^a, c_i^n} \log(c_i^a - \bar{a}) + \nu \log(c_i^n)$$

such that

$$p^a c_i^a + p^n c_i^n \leq y_i + \pi + \tilde{m}$$

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ESTIMATION

- assigned parameters: labor share θ , non-ag cons. weight ν , migration cost disp. σ_m , productivities A_u^n, A_r^a

ESTIMATION

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- estimated parameters:

Parameter	Value	Moment	Data	Model
$\bar{a}$	5.569	food cons. share	0.345	0.345
A_r^n	0.113	median urban-rural inc. gap	5.147	5.138
σ_u	1.547	SD of log urban inc.	1.491	1.492
σ_r	1.982	SD of log rural inc.	1.569	1.569
d	0.075	log urban-rural inc. slope	0.065	0.065
μ_m	2.869	migration rate	0.171	0.171

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μ_m	2.869	migration rate	0.171	0.171

- info friction γ_i :
 - assume common friction: $\gamma_i = \gamma$
 - $\text{avg} \frac{\text{true urban income}}{\text{perceived urban income}} = 0.559$
- $\rightarrow \gamma = \frac{1}{0.559} - 1 = 0.789$

SELECTION

- consider a HH “type” with common $(\gamma_i, m_i, \varepsilon_{r,i}^n, \varepsilon_{r,i}^a)$ but varying $\varepsilon_{u,i}^n$ (and thus $z_{u,i}^n$)

SELECTION

- consider a HH “type” with common $(\gamma_i, m_i, \varepsilon_{r,i}^n, \varepsilon_{r,i}^a)$ but varying $\varepsilon_{u,i}^n$ (and thus $z_{u,i}^n$)
- threshold urban productivity $\hat{z}_{n,i}$ (worker indifferent between migrating/staying):

$$\underbrace{\frac{1}{1+\gamma_i} \frac{1}{1+m_i} \hat{z}_{u,i}^n w_u^n}_{\text{perceived urban inc.}} = \max \left\{ \underbrace{(\hat{z}_{u,i}^n)^d \exp \varepsilon_{r,i}^n w_r^n}_{\text{rural non-ag inc.}}, \underbrace{(\hat{z}_{u,i}^n)^d \exp \varepsilon_{r,i}^a w_r^a}_{\text{rural ag inc.}} \right\}$$

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- if $d < 1$ (rural productivity rises **less** than 1-to-1 with urban productivity):
 - **positive** selection: everyone with $z_{u,i}^n > \hat{z}_{u,i}^n$ migrates
 - $\frac{\partial \hat{z}_{u,i}^n}{\partial \gamma_i} > 0, \frac{\partial \hat{z}_{u,i}^n}{\partial m_i} > 0$ (frictions $\uparrow \rightarrow$ threshold $\uparrow$)

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- If $d > 1$ (rural productivity rises **more** than 1-to-1 with urban productivity):
 - **negative** selection: everyone with $z_{u,i}^n < \hat{z}_{u,i}^n$ migrates
 - $\frac{\partial \hat{z}_{u,i}^n}{\partial \gamma_i} < 0, \frac{\partial \hat{z}_{u,i}^n}{\partial m_i} < 0$ (frictions $\uparrow \rightarrow$ threshold $\downarrow$)

EXPERIMENTAL IMPACTS

Model: Experimental Impacts

	Partial Equilibrium		General Equilibrium		
	Migration rate change	Avg income change	Migration rate change	Avg income change	Avg income change, spillover
Info	0.004	0.113	0.004	0.103	0.001
Mentor	0.010	0.287	0.009	0.272	0.003
Group	0.016	0.155	0.015	0.154	0.006

“Migration rate change” is the difference in migration rate among treated households. “Avg income change” is the average relative change in observed income (gross of migration cost) for the treated households. “Avg income change, spillover” is the average relative change in observed income for the untreated rural households.

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SCALING THE INTERVENTIONS

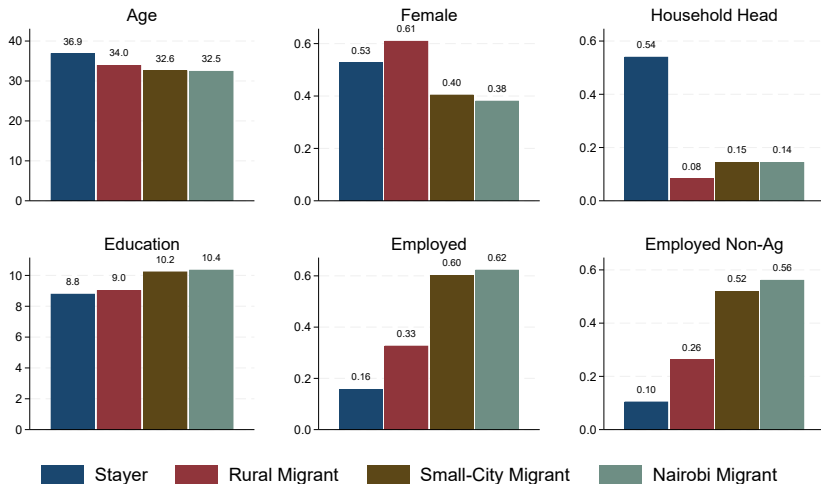
Model: Aggregate Impacts of Universal Treatments

	Migration Rate	Real Non-Agric. GDP	Real Agric. GDP	Agricultural Productivity Gap	Urban-Rural Income Gap
Baseline	0.171	1.000	1.000	2.639	5.138
Info	0.173	1.004	1.000	2.620	5.064
Mentor	0.178	1.013	0.999	2.593	4.962
Perfect Info	0.219	1.060	0.995	2.329	4.086

Universal treatments are applied to all rural households. All economies are solved in general equilibrium. Real non-agricultural and agricultural GDPs are expressed relative to the baseline.

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MIGRANTS ARE MORE EDUCATED, YOUNGER, MALE COMPARED TO RURAL STAYERS



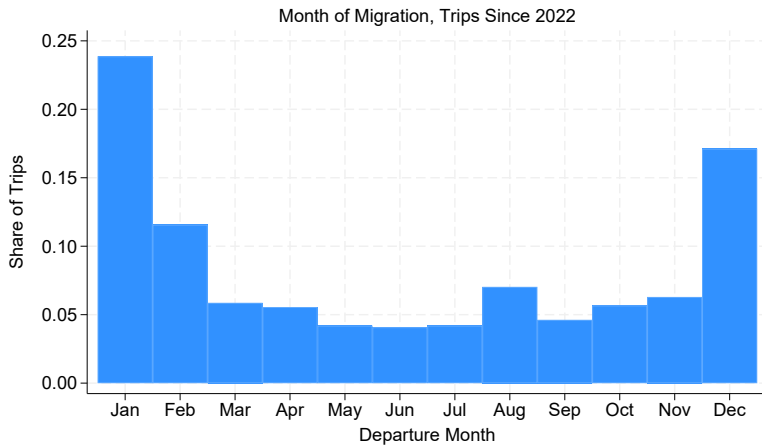
TRIP CHARACTERISTICS

<i>Among those leaving after baseline:</i>	Mean (Control)
Returned to village	0.35
Duration (months)	6.63
Migrated with others from household	0.04
Migrated with others from village	0.03
Received housing assistance from village	0.11
Received job referral from village	0.72
Worked as employee	0.59
Worked as business owner	0.07
Weeks taken to find job after migrating	3.5
Married	0.34

Includes all work migration trips to cities after the baseline survey (measured 16 months later).

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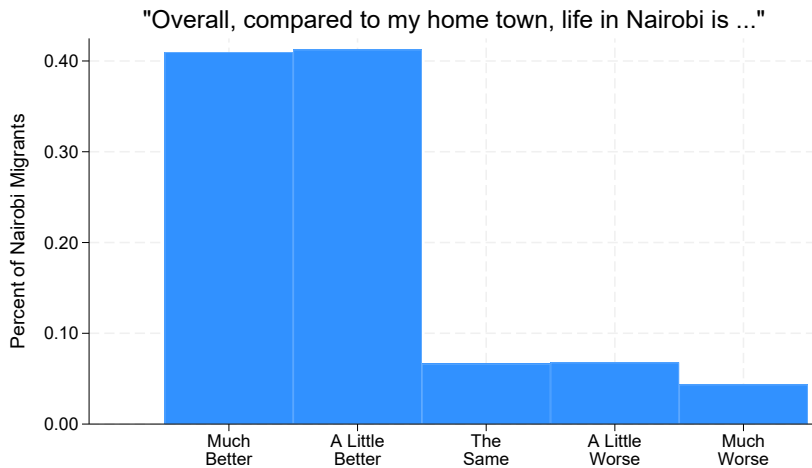
MIGRATION IS PARTLY SEASONAL



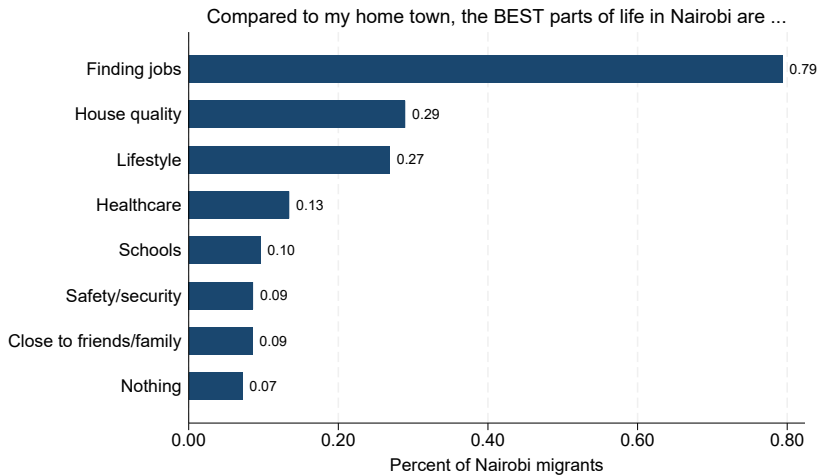
Note: Includes Nairobi migrants only.

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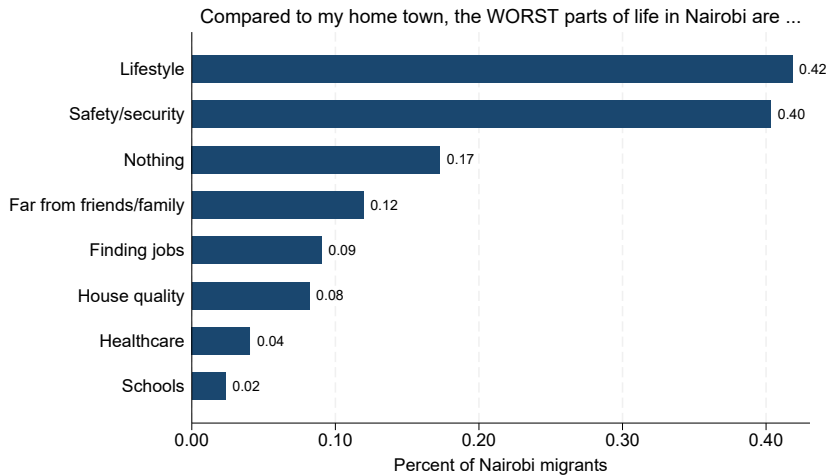
MOST MIGRANTS PREFER LIFE IN THE CITY



MIGRANTS LIKE EASE OF JOB SEARCH, HOUSE QUALITY



BUT DISLIKE SAFETY ISSUES



RURAL HOUSEHOLDS—ESPECIALLY THOSE WITH NO MIGRATION HISTORY—HAVE LIMITED URBAN NETWORKS

	Mean	Coeff. on Experience	p-Value	N
Any member has ever worked in Nairobi	0.44	1.00	.	53,096
Has Nairobi housing network	0.50	0.31	0.00	52,969
Has Nairobi jobs network	0.46	0.29	0.00	52,969
Plans to migrate to Nairobi	0.21	0.14	0.00	53,096
# households in village they give job advice to	1.39	0.51	0.00	52,870

Data from census surveys. *Experience* = 1 if the household has ever had a member working in Nairobi.

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CONSUMER PRICE INDICES

- Following our pre-analysis plan, we construct Laspeyres-style spatial indices:
 - **Nairobi CPI (NCPI)**
 - **Urban CPI excluding Nairobi (UCPI)**
 - Deflate based on location spent (subtract remittances from migrant income)
- Both are defined relative to rural prices using:
 - KIHBS data on quantities and expenditures
 - Our survey data on perceived quality

For each good g in category c (food, rent, HH goods, education, health) in rural areas R and Nairobi N , with data on consumption quantities x and unit prices p :

$$NCPI_c = \frac{\sum_{g \in c} (x_g^R \times p_g^N)}{\sum_{g \in c} (x_g^R \times p_g^R) \times QUAL_c^N}$$

CONSUMER PRICE INDICES (CONT'D)

Location-specific prices are quantity-weighted at the household i level:

$$p_g^L = \frac{\sum_{i \in L} x_{ig} p_{ig}}{\sum_{i \in L} x_{ig}}, \quad L \in \{R, N, T\}$$

Quality adjustment (*QUAL*) from endline survey:

- Ask households how much more they'd pay for urban-quality goods
- Compute household-level ratios, then average

To aggregate across categories:

$$NCPI = \sum_c NCPI_c \times Expenditure_c$$

CONSUMER PRICE INDICES: RESULTS

	NCPI	NCPI	UCPI	UCPI
Food	1.16	0.58	1.10	0.55
Rent	4.22	2.11	1.71	1.02
Household items	1.70	0.85	1.35	0.90
Education	1.86	0.93	1.15	0.86
Healthcare	2.01	1.26	0.95	0.63
Average basket	2.34	1.20	1.31	0.79
Quality-Adjusted?		X		X

NCPI (UCPI) is a Laspeyres-style consumer price index for Nairobi (other urban areas). We quality adjust by dividing nominal price differences by stated willingness to pay for Nairobi (other urban) quality goods (expressed as a multiple of actual expenditure), averaged over former migrant households.

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AMENITY-ADJUSTED INCOME

- Spatial income differences may reflect compensating differentials for amenities. How to value these?
- We elicit migrants' valuations by asking:

“What’s the lowest income per month that would convince you to move back to your hometown, if you could earn it while living there?”

We then compute amenity-adjusted family income as:

$$\text{Income}_i = \text{Village Profit}_i + \sum_{m \in i} \text{Rural Indifference Income}_{im}$$

*Rural Indifference Income*_{im} = elicited amount (for migrants) or actual income (for non-migrants)

AMENITY-ADJUSTED INCOME: VALIDATION

	Outcome: Rural indifference income				
Earned urban income	0.21*** (0.01)	0.21*** (0.01)	0.21*** (0.01)	0.19*** (0.02)	0.16*** (0.02)
Prefer life in Nairobi to hometown		0.07** (0.03)	0.06** (0.03)	0.04 (0.03)	0.04 (0.03)
Happy with life			0.11*** (0.03)	0.08*** (0.03)	0.07** (0.03)
Utility quality index (standardized)				0.05*** (0.02)	0.06*** (0.02)
Intended duration in Nairobi (standardized)				0.05*** (0.02)	0.05*** (0.02)
Demographic Controls					X
Observations	1,056	1,052	1,052	1,033	1,029

Note: All columns use Poisson regression. Outcome mean = 120 USD/month.

MIGRANTS REPORT BEING WILLING TO MOVE HOME ONLY FOR HIGHER INCOMES

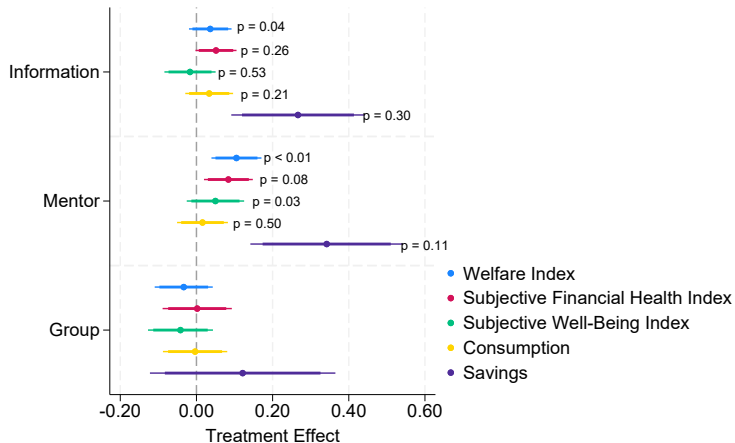


Correlation coefficient = 0.42. Bandwidth = 20. Incomes are top-coded at 230 USD.

Data from endline phone surveys of migrants living in Nairobi. *Rural indifference income* is the lowest reported income the migrant would be willing to move back to their hometown for.

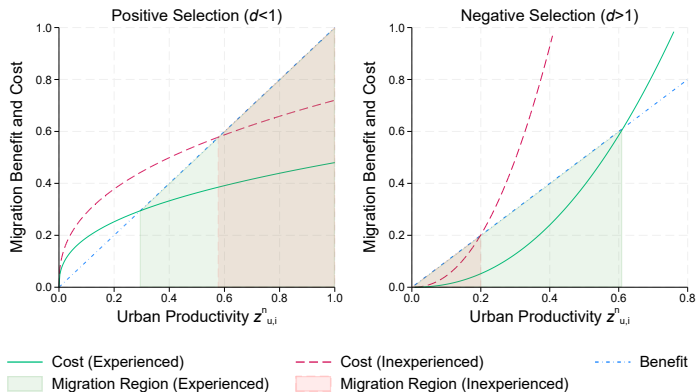
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OTHER ECONOMIC IMPACTS



Units are standard deviations (index) or % changes. p -values on chart test equivalence to **group**.

SELECTION AND RETURNS



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